



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

Full Syllabus

26-07-2021

## JEE(Main + Advanced) : Enthusiast Course (ALL STAR BATCHES)

Test Type : SCORE-III

Test Pattern : JEE (Main)

### TEST DATE : 17-07-2021

#### ANSWER KEY

##### PHYSICS

| SEC-1 | Q. | 1      | 2            | 3    | 4    | 5     | 6     | 7     | 8    | 9    | 10     |
|-------|----|--------|--------------|------|------|-------|-------|-------|------|------|--------|
|       | A. | B      | D            | B    | A    | B     | D     | C     | C    | B    | D      |
| SEC-1 | Q. | 11     | 12           | 13   | 14   | 15    | 16    | 17    | 18   | 19   | 20     |
|       | A. | A      | B            | D    | D    | A     | C     | C     | B    | A    | B      |
| SEC-2 | Q. | 1      | 2            | 3    | 4    | 5     | 6     | 7     | 8    | 9    | 10     |
|       | A. | 360.00 | 0.66 to 0.67 | 2.50 | 4.00 | 40.00 | 10.00 | 40.00 | 2.00 | 2.00 | 400.00 |

##### CHEMISTRY

| SEC-1 | Q. | 1    | 2    | 3     | 4     | 5      | 6       | 7    | 8    | 9      | 10   |
|-------|----|------|------|-------|-------|--------|---------|------|------|--------|------|
|       | A. | B    | A    | D     | D     | B      | C       | A    | C    | D      | D    |
| SEC-1 | Q. | 11   | 12   | 13    | 14    | 15     | 16      | 17   | 18   | 19     | 20   |
|       | A. | A    | A    | B     | C     | B      | A       | C    | D    | A      | A    |
| SEC-2 | Q. | 1    | 2    | 3     | 4     | 5      | 6       | 7    | 8    | 9      | 10   |
|       | A. | 0.50 | 3.00 | 16.00 | 40.00 | 211.50 | 1000.00 | 8.00 | 5.00 | 100.10 | 0.36 |

##### MATHEMATICS

| SEC-1 | Q. | 1     | 2    | 3    | 4    | 5    | 6     | 7    | 8    | 9    | 10   |
|-------|----|-------|------|------|------|------|-------|------|------|------|------|
|       | A. | D     | C    | D    | C    | D    | A     | C    | A    | B    | A    |
| SEC-1 | Q. | 11    | 12   | 13   | 14   | 15   | 16    | 17   | 18   | 19   | 20   |
|       | A. | A     | C    | C    | A    | D    | B     | D    | A    | C    | A    |
| SEC-2 | Q. | 1     | 2    | 3    | 4    | 5    | 6     | 7    | 8    | 9    | 10   |
|       | A. | 99.75 | 2.00 | 1.00 | 1.00 | 0.50 | 15.00 | 6.75 | 9.00 | 1.50 | 8.50 |

#### HINT – SHEET

##### PHYSICS

###### SEC-1

1. **Ans ( B )**

$$I = \frac{5}{4}MR^2$$

$$\frac{dI}{I} = \frac{dM}{M} + 2\frac{dR}{R}$$

$$= 1\% + 2(1.5\%) = 1\% + 3\% = 4\%$$

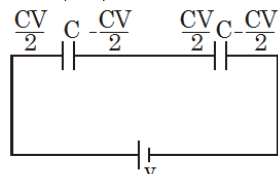
2. **Ans ( D )**

Magnetic field cannot change speed and since magnetic field is along z axis, velocity cannot be in z axis at any instant i.e.. it cannot have  $\hat{k}$  component.

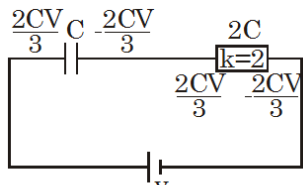
3. **Ans ( B )**

$$V_A = \frac{kQ}{R} + k\frac{2Q}{2R} + k\frac{8Q}{4R} = \frac{4kQ}{R}$$

4. **Ans (A)**



initially



finally

$$\text{charge given by battery} = \frac{2CV}{3} - \frac{CV}{2}$$

$$= \frac{CV}{6}$$

$$\Rightarrow \text{work done by battery} = \frac{CV^2}{6}$$

Initial energy stored

$$= \left( \frac{1}{2} C \left( \frac{V}{2} \right)^2 \right) \times 2 = \frac{CV^2}{4}$$

Final energy stored

$$= \frac{1}{2} C \left( \frac{2V}{3} \right)^2 + \frac{1}{2} 2C \left( \frac{V}{3} \right)^2 = \frac{CV^2}{3}$$

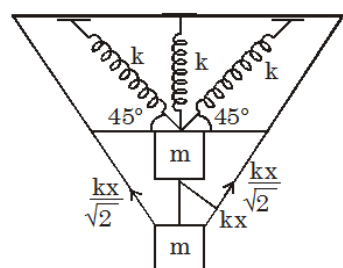
$$\Rightarrow \Delta U = \frac{CV^2}{3} - \frac{CV^2}{4} = \frac{CV^2}{12}$$

$W_b = \Delta U + \text{Heat loss}$

$$\frac{CV^2}{6} = \frac{CV^2}{12} + \text{Heat loss}$$

$$\Rightarrow \text{Heat loss} = \frac{CV^2}{12}$$

5. **Ans (B)**



Net vertical force

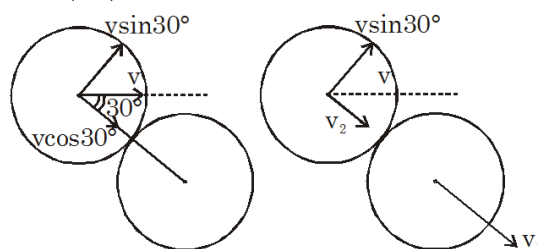
$$= kx + \frac{kx}{\sqrt{2}} \cos 45^\circ \times 2 = 2kx = ma$$

$$a = \frac{2k}{m} x$$

$$\Rightarrow \omega = \sqrt{\frac{2k}{m}}$$

$$\tau = 2\pi \sqrt{\frac{m}{2k}}$$

6. **Ans (D)**



$$mv \cos 30^\circ = mv_2 + 2mv_1$$

$$e = \frac{1}{2} = \frac{v_1 - v_2}{v \cos 30^\circ}$$

$$\text{on solving } v_1 = \frac{\sqrt{3}v}{4}$$

8. **Ans (C)**

$$i_s R_s = i_g R_g$$

$$\frac{20 \times 50}{1000} = i_s \times 0.01$$

$$i_s = 100 \text{ A}$$

9. **Ans (B)**

$$V_{\max} = V_m + V_c$$

$$V_{\min} = V_c - V_m$$

$$\text{modulation index} = \frac{V_m}{V_c} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

10. **Ans (D)**

$$\text{After passing through first } I = \frac{I_0}{2}$$

$$\frac{I_0}{4} = \frac{I_0}{2} \cos^2 \phi$$

$$\Rightarrow \cos^2 \phi = \frac{1}{2}$$

$$\cos \phi = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \phi = 45^\circ$$

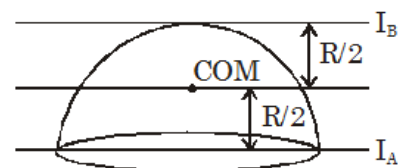
11. **Ans (A)**

$$\text{Initially } \frac{R_1}{R_2} = \frac{2}{3} = \frac{\ell}{100 - \ell} \Rightarrow \ell = 40 \text{ cm}$$

$$\text{Now } \frac{R_2}{R_1} = \frac{3}{2} = \frac{\ell}{100 - \ell} \Rightarrow \ell = 60 \text{ cm}$$

$$\text{shift} = 20 \text{ cm}$$

12. **Ans (B)**



$$I_A = I_B = I_{\text{CM}} + M \left( \frac{R}{2} \right)^2$$

13. **Ans (D)**

Since initial momentum is zero. So final momentum is also zero. So momentum of hydrogen atom and photon will be equal.

14. **Ans (D)**

Resolving power  $\propto \frac{a}{\lambda f}$

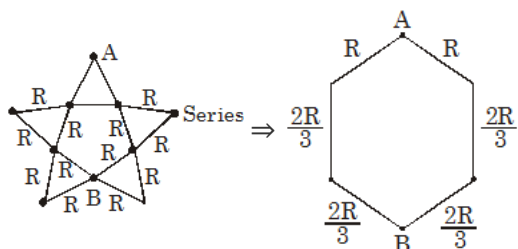
$a$  : aperture

$\lambda$  : wavelength

$f$  : focal length

By immersing objective lens in oil, resolving power increases.

15. **Ans (A)**



16. **Ans (C)**

$$KE \propto e^2 \cdot (Ze)^2$$

if  $e$  alone is doubled, charge on nucleus

i.e.  $(Ze)$  will remain constant

$\Rightarrow$  K.E. becomes 4 times.

17. **Ans (C)**

$$TP^{-1/3} = \text{constant}$$

$$PVP^{-1/3} = \text{constant}$$

$$PV^{3/2} = \text{constant}$$

$$\Rightarrow x = \frac{3}{2}, \gamma = \frac{5}{3}$$

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - x} = \frac{3R}{2} + \frac{R}{1 - \frac{3}{2}} = -\frac{R}{2}$$

$\therefore C$  is negative

$$\Delta Q = nC\Delta T$$

$\Rightarrow$  if  $\Delta Q$  is +ve,  $\Delta T$  is -ve

$\Rightarrow$  temperature will decrease

18. **Ans (B)**

If length halved, area is doubled

$$\Rightarrow \text{Breaking stress} = \frac{W}{A} = \frac{W'}{2A}$$

to break new wire load  $W' = 2W$

on applying  $W$ , new wire will not break.

19. **Ans (A)**

Momentum is conserved in all types of collision.

**PHYSICS**

**SEC-2**

1. **Ans (360.00)**

In diffraction pattern

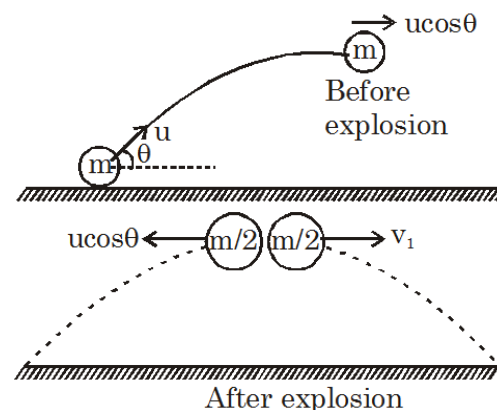
$$\text{First minima} = \frac{\lambda D}{a}$$

$$\text{First maxima} = \frac{3\lambda D}{2a}$$

$$\Rightarrow \frac{3\lambda D}{2a} = \frac{(540) D}{a}$$

$$\Rightarrow \lambda = 360 \text{ nm}$$

2. **Ans (0.66 to 0.67)**



from momentum conservation

$$mu \cos \theta = -\frac{m}{2} u \cos \theta + \frac{m}{2} v_1$$

$$v_1 = 3u \cos \theta$$

$$\lambda_D = \frac{h}{mv}$$

$$\Rightarrow \frac{(\lambda_D)}{(\lambda_D)_2} = \frac{\frac{h}{\frac{m}{2}(3u \cos \theta)}}{\frac{h}{mu \cos \theta}} = \frac{2}{3}$$

3. **Ans (2.50)**

$$V_{BE} = 0.7V = 8.7 - I_B(200 \times 10^3)$$

$$\Rightarrow I_B = \frac{8}{200 \times 10^3} = 4 \times 10^{-5} A$$

$$\Rightarrow \beta = 80 = \frac{I_C}{I_B} \Rightarrow I_C = 32 \times 10^{-4} A$$

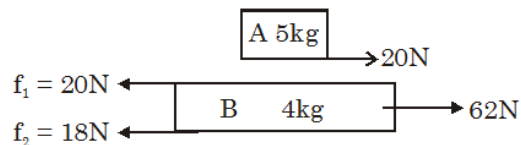
$$V_{CE} = 3V = 11 - I_C R_C$$

$$3 = 11 - 32 \times 10^{-4} R_C$$

$$\Rightarrow R_C = \frac{8}{32 \times 10^{-4}}$$

$$R_C = 2.5 \text{ k}\Omega$$

4. **Ans ( 4.00 )**



$$\Rightarrow a_A = 4 \text{ m/s}^2 \text{ (Right)}$$

$$a_B = 6 \text{ m/s}^2 \text{ (Right)}$$

$$a_{A/B} = 2 \text{ m/s}^2 \text{ towards left}$$

$$\Rightarrow 16 = \frac{1}{2} 2 t^2 \Rightarrow t = 4 \text{ sec}$$

5. **Ans ( 40.00 )**

velocity when string becomes horizontal

$$v = \sqrt{u^2 - 2gl} = \sqrt{60 - 20}$$

$$v = \sqrt{40}$$

$$\Rightarrow T = \frac{mv^2}{l} = \frac{1(40)}{1}$$

$$\Rightarrow T = 40 \text{ N}$$

6. **Ans ( 10.00 )**

Powerless across circuit is maximum when

$$R = X_L = 10 \Omega$$

$$P_{\max} = \frac{V_{\text{rms}}^2}{2R} = \frac{V_0^2}{4R} = \frac{(20)^2}{4 \times 10} = 10 \text{ W}$$

7. **Ans ( 40.00 )**

$$\text{Stopping distance} = \frac{u^2}{2a} = \frac{(20)^2}{2(5)} = 40 \text{ m}$$

8. **Ans ( 2.00 )**

$$v = 8 \text{ m/s} = R\omega$$

$$\Rightarrow \omega = 4 = t^2 - 4t + 8$$

$$t^2 - 4t + 4 = 0 \Rightarrow t = 2 \text{ sec}$$

9. **Ans ( 2.00 )**

$$R = 50 \Omega$$

$$X_L = \omega L = 50 \Omega$$

$$X_L = \frac{1}{\omega C} = 100 \Omega$$

$$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = 50 \sqrt{2} \Omega$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200/\sqrt{2}}{50\sqrt{2}} = 2 \text{ A}$$

10. **Ans ( 400.00 )**

III<sup>rd</sup> overtone

$$f = \frac{4V}{2L} = \frac{2V}{L} = \frac{2}{1} \sqrt{\frac{200}{(5/1000)}} = 2(200)$$

$$f = 400 \text{ Hz}$$

## CHEMISTRY

### SEC-1

2. **Ans ( A )**

$$\text{N} \Rightarrow 1s^2 2s^2 2p^3 \Rightarrow \text{N}^+ \Rightarrow 1s^2 2s^2 2p^2$$

$$\text{O} = 1s^2 2s^2 2p^4 \Rightarrow \text{O}^+ \Rightarrow 1s^2 2s^2 2p^3$$

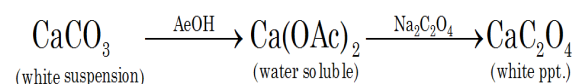
$$\text{NO} \Rightarrow \sigma_{1s}^2, \sigma_{1s}^{*2}, \sigma_{2s}^2, \sigma_{2s}^{*2}, \pi_{2py}^2 = \pi_{2pz}^2, \sigma_{2px}^2, \pi_{2py}^1 = \pi_{2pz}^0$$

8. **Ans ( C )**

Van-Arkel process  $\Rightarrow$  Ti, Hf, Zr, B

Mond's process  $\Rightarrow$  Ni

10. **Ans ( D )**



12. **Ans ( A )**

(1) In f-block on moving from left to right due to lanthanide contraction size decreases, hence E.N. increases so acidic nature of hydroxides increases.

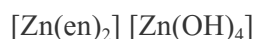
(2)  $\text{C}_2^{2-}$  on hydrolysis gives  $\text{C}_2\text{H}_2$

(3) & (4) f-block elements uses

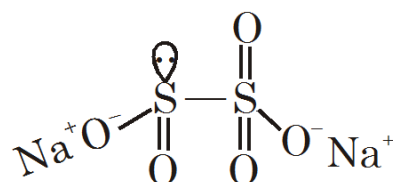
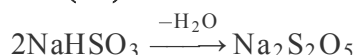
16. **Ans ( A )**

After sharing all valence electrons, when central atom completes its octet (without any lone pairs), then it is called electron precise hydride.

18. **Ans ( D )**



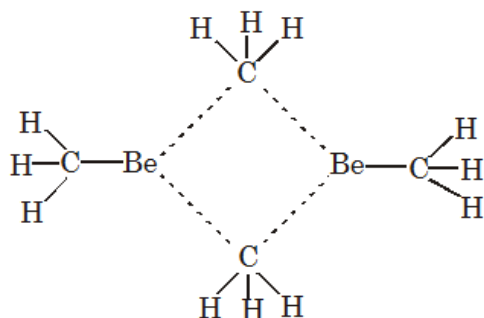
20. **Ans ( A )**



## CHEMISTRY

### SEC-2

3. Ans ( 16.00 )



$$3c - 2e = 2$$

$$2c - 2e = 14$$

$$X = 2$$

$$Y = 14$$

$$X + Y = 16$$

8. Ans ( 5.00 )

Sudden Jump is being observed after  $IE_5$   
 $|IE_6 - IE_5|$  difference is maximum than any  
 other successive IE difference

Hence valence  $e^- = 5$ .

## MATHEMATICS

### SEC-1

2. Ans ( C )

$$x^2 - 2x - a^2 + 1 = 0 \Rightarrow (x - 1)^2 - a^2 = 0$$

$$\Rightarrow x = 1 \pm a$$

$$\text{Let } f(x) = x^2 - 2(a + 1)x + a(a - 1)$$

$$f(a + 1) < 0$$

$$\Rightarrow (a + 1)^2 - 2(a + 1)^2 + a(a - 1) < 0$$

$$\Rightarrow -a^2 - 2a - 1 + a^2 - a < 0$$

$$\Rightarrow a > -\frac{1}{3} \quad \dots(1)$$

$$f(1 - a) < 0$$

$$(1 - a)^2 - 2(a + 1)(1 - a) + a(a - 1) < 0$$

$$\Rightarrow (a - 1)[a - 1 + 2a + 2 + a] < 0$$

$$\Rightarrow (a - 1)(4a + 1) < 0$$

$$\Rightarrow a \in \left(-\frac{1}{4}, 1\right) \quad \dots(2)$$

$$\text{From (1), (2)} \Rightarrow a \in \left(-\frac{1}{4}, 1\right)$$

3. Ans ( D )

$$\text{As } AA^T = I_n \Rightarrow A \text{ is orthogonal}$$

$$\Rightarrow \det A \neq 0 \Rightarrow A \text{ is non-singular.}$$

4. Ans ( C )

$$\Delta = \begin{vmatrix} 1 & 1 & -a \\ 2 & a & 1 \\ a & 1 & -1 \end{vmatrix} = 1(-a - 1) - 1(-2 - a) - a(2 - a^2)$$

$$= a^3 - 2a + 1 = (a - 1)(a^2 + a - 1)$$

$$= (a - 1) \left( a - \frac{-1 - \sqrt{5}}{2} \right) \left( a - \frac{-1 + \sqrt{5}}{2} \right)$$

$$\text{For } \Delta = 0$$

$$a = 1, \frac{-1 \pm \sqrt{5}}{2}$$

for each value system has no solution.

5. Ans ( D )

$$\sum_{0 \leq i < j \leq n} i \binom{n}{j} = \frac{n(n-1)}{2} \sum_{k=0}^n \binom{n-2}{k}$$

$$= \frac{n(n-1)}{2} 2^{n-2} = n(n-1)2^{n-3}$$

6. Ans ( A )

$$S_k = \sum a_1 a_2 \dots a_k \quad \text{and}$$

$$S_{n-k} = \sum a_1 a_2 \dots a_{n-k}$$

$$= a_1 a_2 \dots a_n \sum \frac{1}{a_1 a_2 \dots a_k}$$

Now apply  $AM \geq HM$

$$\frac{\sum a_1 a_2 \dots a_k}{\binom{n}{k}} \geq \frac{\binom{n}{k}}{\sum \frac{1}{a_1 a_2 \dots a_k}}$$

$$\Rightarrow \frac{S_k}{\binom{n}{k}} \geq \frac{\binom{n}{k}}{S_{n-k}} \quad (a_1 a_2 \dots a_n)$$

$$\Rightarrow S_k S_{n-k} \geq \binom{n}{k}^2 (a_1 a_2 \dots a_n)$$

7. Ans ( C )

$$T_r = \frac{\tan 2^{r-1}}{\cos 2^r} = \tan 2^r - \tan 2^{r-1}$$

$$\Rightarrow \sum_{r=1}^{100} T_r = \tan 2^{100} - \tan 1$$

8. **Ans (A)**

$$f(x) = \begin{cases} 0, & x \in [0, 1) \\ x, & x \in (1, 2] \\ 2(x-1), & x \in (2, 3) \end{cases}$$

$f(x)$  is discontinuous at  $x = 1$ , hence it is not differentiable at that point.

At  $x = 2$

LHD = 1 and RHD = 2

$\Rightarrow f(x)$  is not differentiable at  $x = 2$ .

9. **Ans (B)**

Given  $2x^2 + y^2 = 12$

$$\text{So, } 4x + 2yy' = 0 \Rightarrow y' = -\frac{2x}{y}$$

$\Rightarrow$  Slope of tangent = -2

$\Rightarrow$  Slope of normal =  $\frac{1}{2}$

Hence, equation of normal is  $2y = x + 2$

Solving with the ellipse

$$x = \frac{-22}{9} \text{ and } y = \frac{-2}{9}$$

10. **Ans (A)**

$$\int e^{x^2} \left( 2 - \frac{1}{x^2} \right) dx = \int e^{x^2} \left( 2x \cdot \frac{1}{x} - \frac{1}{x^2} \right) dx$$

$$= \frac{e^{x^2}}{x} + C$$

$$\text{So, } f(x) = \frac{1}{x}$$

11. **Ans (A)**

$$\frac{2b^2}{a} = 2ae \Rightarrow a^2(1 - e^2) = a^2e$$

$$\Rightarrow e^2 + e - 1 = 0 \Rightarrow e = \frac{\sqrt{5} - 1}{2}$$

$$= 2\sin 18^\circ$$

12. **Ans (C)**

$$\frac{1}{y(\ln y)^2} \frac{dy}{dx} + \frac{1}{\ln y} \times \frac{1}{x} = \frac{1}{x^2}$$

$$\text{Put } \frac{1}{\ln y} = t, \text{ So, } \frac{-1}{y(\ln y)^2} \frac{dy}{dx} = \frac{dt}{dx}$$

Hence, DE becomes

$$\frac{dt}{dx} - \frac{t}{x} = \frac{-1}{x^2} \Rightarrow \text{IF} = \frac{1}{x}$$

$$\text{So, } \frac{t}{x} = - \int \frac{1}{x^3} dx + C = \frac{1}{2x^2} + C$$

$$\Rightarrow \frac{1}{\ln y} = \frac{1}{2x} + Cx$$

13. **Ans (C)**

$$C_1(0, 0) \text{ and } r_1 = 2$$

$$\text{Also, } C_2(4, 4) \text{ and } r_2 = 5$$

$$\text{Since, } C_1C_2 = 4\sqrt{2}$$

$$\Rightarrow |r_1 - r_2| < C_1C_2 < r_1 + r_2$$

$\Rightarrow$  Circles are intersecting.

$\Rightarrow$  Two direct common tangents exist.

14. **Ans (A)**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{Slope of normal is } -\frac{a^2}{2b^2}$$

$\therefore$  Equation of normal at  $P(6, 3)$  is

$$y - 3 = -\frac{a^2}{2b^2}(x - 6)$$

$$\text{at } (10, 0) \Rightarrow -3 = -\frac{a^2}{2b^2}(10 - 6)$$

$$\Rightarrow \frac{3}{2} = \frac{a^2}{b^2}$$

$$\therefore e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{2}{3} = \frac{5}{3}$$

$$\therefore e = \sqrt{\frac{5}{3}}$$

15. **Ans (D)**

$$4 + \alpha + 2 = 0 \Rightarrow \alpha = -6$$

$$\text{Also, } x = 2\lambda + 1, y = \alpha\lambda - \alpha = -6\lambda + 6 \text{ and } z =$$

$$2\lambda + \beta$$

$$\text{lies on the plane } 2x + y + z = 5$$

$$\Rightarrow 4\lambda + 2 - 6\lambda + 6 + 2\lambda + \beta = 5$$

$$\Rightarrow \beta = -3$$

$$\text{So } \alpha + \beta = -9$$

16. **Ans (B)**

Family of plane is

$$(2x + 3y + 4z + 5) + \lambda(x + y + z - 5) = 0$$

$$\Rightarrow (2 + \lambda)x + (3 + \lambda)y + (4 + \lambda)z + (5 - 5\lambda) = 0$$

$$\text{So } 2 + \lambda + 3 + \lambda + 4 + \lambda = 0 \Rightarrow \lambda = -3$$

$$\text{So, required plane is } -x + z + 20 = 0$$

$$x - z = 20$$

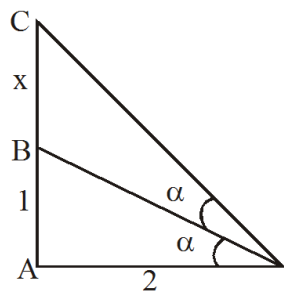
17. Ans (D)

Required probability

$$\frac{{}^{13}C_3}{{}^{15}C_3} = \frac{\frac{13 \times 12 \times 11 \times 10!}{3! \times 10!}}{\frac{15 \times 14 \times 13 \times 12!}{3! \times 12!}}$$

$$= \frac{13 \times 12 \times 11}{15 \times 14 \times 13} = \frac{12 \times 11}{15 \times 14} = \frac{22}{35}$$

18. Ans (A)



$$\tan \alpha = \frac{1}{2}$$

$$\tan 2\alpha = \frac{1+x}{2}$$

$$= \frac{2 \times \frac{1}{2}}{1 - \frac{1}{4}} = \frac{1+x}{2}$$

$$= \frac{1 \times 4}{3} = \frac{1+x}{2}$$

$$\Rightarrow 1+x = \frac{8}{3} \Rightarrow x = \frac{5}{3}$$

19. Ans (C)

$$x > 0$$

$$\cos^{-1}x + \cos^{-1}2x = \pi$$

$$2x = \cos(\pi - \cos^{-1}x)$$

$$= -\cos(\cos^{-1}x)$$

$$= -x \Rightarrow x = 0$$

$$x < 0$$

$$\cos^{-1}(-x) + \cos^{-1}(-2x) = \pi$$

$$\pi - \cos^{-1}x + \pi - \cos^{-1}(2x) = \pi$$

$$\cos^{-1}x + \cos^{-1}(2x) = \pi$$

20. Ans (A)

Negation of  $\sim p \vee (p \vee (\sim q))$  is

$$p \wedge \sim(p \vee \sim q)$$

$$\equiv p \wedge \sim q \wedge q$$

$$\equiv (\sim p \wedge q) \wedge p$$

MATHEMATICS

SEC-2

1. Ans (99.75)

$$|(a+ib)z - z| = |(a+ib)z|$$

$$|z||a-1+ib| = |z||a+ib|$$

$$\Rightarrow a = \frac{1}{2}$$

$$\text{Now } a^2 + b^2 = 100$$

$$\Rightarrow b^2 = \frac{399}{4}$$

2. Ans (2.00)

$$2B + A = A^T - B^T$$

$$\Rightarrow 2B^T + A^T = A - B$$

$$\Rightarrow 3B + 3B^T = 0$$

$$\Rightarrow 2B + A = A^T + B$$

$$\Rightarrow B + A = A^T$$

$$\Rightarrow |B + A| = |A^T| = |A| = 2$$

3. Ans (1.00)

$$\lim_{x \rightarrow \infty} \left( \frac{a^{1/x} + b}{c} \right)^x = \begin{cases} 0, & c > b+1 \\ a^{1/c}, & c = b+1 \\ \infty, & c < b+1 \end{cases}$$

$$\Rightarrow (b+1)\log_a d = 1$$

5. Ans (0.50)

Second integral vanishes

or its integrand is odd function.

$$\text{Now, } \cos x + \cos 3x + \cos 5x + \cos 7x$$

$$= \frac{\cos 4x \sin 4x}{\sin x}$$

$$\text{So, } \left( \frac{\sin 8x}{\sin x} \right) \cos x = (1 + \cos 2x)$$

$$+ (\cos 4x + \cos 2x) + (\cos 6x + \cos 4x)$$

$$+ (\cos 8x + \cos 6x)$$

$$\text{Hence, } \int_0^{\pi/2} \sin 8x \cot x dx = \frac{\pi}{2}$$

6. Ans (15.00)

Required number of tetrahedrons

$$= {}^3C_3 \times {}^3C_1 + {}^3C_2 \times {}^3C_2 + {}^3C_1 \times {}^3C_3 = 15$$

7. Ans ( 6.75 )

Put  $y = 9$  in given pair of straight lines

$$18x^2 - 81x + 81 = 0$$

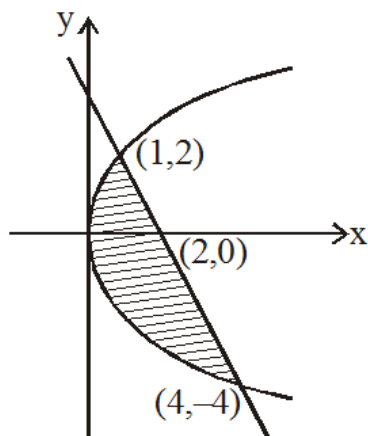
$$\Rightarrow x = 3, \frac{3}{2}$$

So, we have three vertices  $(0, 0)$ ,  $(3, 9)$  and

$$\left(\frac{3}{2}, 9\right)$$

$$\text{Hence } \Delta = \frac{27}{4}$$

8. Ans ( 9.00 )



Required area

$$= \int_{-4}^2 \left( \frac{4-y}{2} - \frac{y^2}{4} \right) dy$$

$$= \left( 2y - \frac{y^2}{4} - \frac{y^3}{12} \right)_{-4}^2$$

$$= 2(2+4) - \frac{1}{4}(4-16) - \frac{1}{12}(8+64)$$

$$= 12 - (-3) - 6 = 9 \text{ sq. units}$$

9. Ans ( 1.50 )

$$|\vec{c} - \vec{a}| = \sqrt{14}$$

$$\Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 14 \dots (1)$$

$$\vec{a} \cdot \vec{c} + 2|\vec{c}| = 0$$

$$\Rightarrow |\vec{a}| \cdot |\vec{c}| \cdot \cos \theta + 2|\vec{c}| = 0$$

$$\Rightarrow |\vec{c}| \cdot (|\vec{a}| \cdot \cos \theta + 2) = 0$$

$$\Rightarrow \cos \theta = -\frac{2}{3} \text{ given } |\vec{a}| = 3$$

from (i)

$$\Rightarrow |\vec{c}|^2 + 9 - 2|\vec{c}| \cdot |\vec{a}| \cdot \left(-\frac{2}{3}\right) - 14 = 0$$

$$\Rightarrow |\vec{c}|^2 + 4|\vec{c}| - 5 = 0 \Rightarrow |\vec{c}| = 1, -5$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = |(\vec{a} \times \vec{b})| \cdot |\vec{c}| \cdot \sin \theta$$

$$= 3 \cdot 1 \times \frac{1}{2} = \frac{3}{2}$$

10. Ans ( 8.50 )

New mean

$$= \frac{x_1 + 1 + x_2 + 2 + x_3 + 3 + x_4 + 4 + \dots + x_{10} + 10}{10}$$

$$= \frac{\sum x_i + \frac{10 \times 11}{2}}{10} = 3 + \frac{11}{2} = \frac{17}{2}$$

